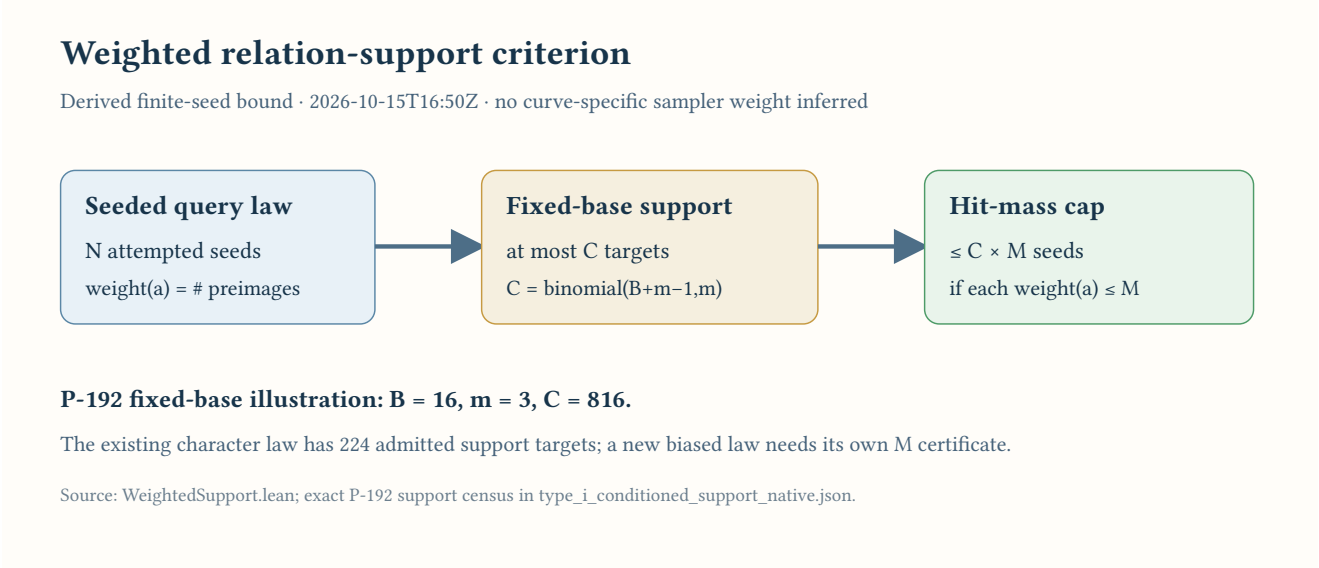
Weighted relation support for geometry-guided curve queries

A finite-seed nonuniform query law can place at most $\binom{B+m-1}{m}M$ seeds on the unordered positive m-sum support of a fixed base of B usable classes, if each accepted target has at most M preimage seeds. Lean checks the full-support and accepted-set bounds.



Proof and probability interpretation

Let $w(a)$ count seeds mapping to target class a. The fixed-base support S has at most $C = \binom{B+m-1}{m}$ classes. On an accepted set E, $\sum_{a \in E \cap S} w(a) \le |E \cap S| M \le CM$. Divide by the attempted seed count for a per-attempt bound, or by accepted seed mass for a conditional bound. The theorem requires an exact maximum-preimage certificate and no uniform-target premise.

Curve-screening consequence

For the P-192 Type I 16-class, three-sum base, $C = 816$. Its earlier character law admits 224 support targets and has a separately certified uniform accepted-target hit bound below 2^{-182}. The new result does not assign a preimage cap to a different sampler; that cap, verified relation yield, useful rank, and target descent are concrete gates for each guided collector.

The source-linked report is `research/weak_curve_leaderboard_20261012/reports/run-20261015T1650Z/REPORT.md`. Lean module: `WeightedSupport.lean` (Lean 4.34.1, pinned Mathlib). The SVG is the editable diagram source.